A Weight-5 Apéry-like Sequence $S_{20}(n)$: Formally Verified Supercongruences and a Conjectural Calabi–Yau Period

(preprint, v7 — under community review)

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Abstract

We study the integer sequence

$$S_{20}(n) \;=\; \sum_{k=0}^{n} \binom{n}{k}^4 \binom{n+k}{k} \;=\; 1,\, 3,\, 55,\, 1155,\, 29751,\, 852753, \ldots,$$

a weight-5 (more precisely weight-$(4,1)$) Apéry-like binomial sum that we have been unable to locate in the literature or in the OEIS. A draft OEIS submission (proposed identifier A397213) is *pending editorial review and has not been approved*; we therefore make no novelty claim beyond "we did not find it." The purpose of this note is narrow and, we hope, honest: we give complete, human-readable proofs of three arithmetic congruences satisfied by S_{20}, and we report that *each of these three results has also been verified in the Lean 4 proof assistant with no `sorry`, no `admit`, and no axioms beyond Lean's three standard foundations* (`propext`, `Classical.choice`, `Quot.sound`). The three results are:

(I) the *cubic supercongruence* $S_{20}(p) \equiv 3 \pmod{p^3}$ for every prime $p \geq 5$;

(II) *Wolstenholme's theorem* $\binom{2p}{p} \equiv 2 \pmod{p^3}$ for $p \geq 5$, reproved from first principles (it is the one external input of (I));

(III) the *Apéry-style boundary congruence* $S_{20}(p-1) \equiv 1 \pmod{p}$ for every prime p.

We are explicit about what is *not* settled: the order-4 minimal linear recurrence is computed but not formally certified for general n; the mod-p^3 refinement of (III) is an open conjecture; and the interpretation of S_{20} as a Calabi–Yau period is a structural analogy, not a theorem. We claim no discovery: S_{20} is one member of a much-studied family, and we publish this as an exploratory preprint expressly to seek skeptical review from specialists in Apéry-like sequences and Calabi–Yau differential operators. Corrections, refutations, a prior appearance of the sequence, and independent verification are all warmly welcomed.

Contents

1 Introduction and honest scope

Apéry's irrationality proofs for $\zeta(2)$ and $\zeta(3)$ rest on integer sequences satisfying three-term recurrences with remarkable arithmetic properties. A now-classical theme — developed by Beukers, Zagier, Almkvist–Zudilin, and others — is that such sequences often coincide with periods of one-parameter families of Calabi–Yau manifolds, and that their p-adic behaviour (supercongruences) reflects this geometry.

This note concerns the specific binomial sum

$$S_{20}(n) \;=\; \sum_{k=0}^{n} \binom{n}{k}^4 \binom{n+k}{k}. \tag{1}$$

It is the $A = 4,\ B = 1$ member of the family $\sum_k \binom{n}{k}^A \binom{n+k}{k}^B$; the Apéry numbers for $\zeta(3)$ are $A = B = 2$.

What this paper claims, and what it does not. We have tried to separate rigour from speculation cleanly. The reader should read the following table as the precise scope of the work.

Statement	Status	Where
First terms / closed form (1)	verified (exact arithmetic)	§2
$S_{20}(p) \equiv 3\ (p^3)$, $p \geq 5$	**proved** (+ Lean)	§3
$\binom{2p}{p} \equiv 2\ (p^3)$, $p \geq 5$	**proved** (+ Lean)	§4
$S_{20}(p-1) \equiv 1\ (p)$, all p	**proved** (+ Lean)	§5
$S_{20}(p-1) \equiv 1\ (p^3)$	open conjecture (numeric only)	§5
order-4 recurrence (general n)	**proved** (Zeilberger certificate)	§7
minimal ODE order of $f(z)$ is 6	computed (exact)	§7
order-4 MUM block (CY 3-fold)	evidence (indicial eq.)	§7
Calabi–Yau period interpretation	conjectural analogy	§7, §8

On the Lean 4 formalization. All three congruence theorems below were transcribed into Lean 4 (Mathlib, toolchain `v4.31.0`). Building the development with `lake build` succeeds with no errors; the command `#print axioms` applied to each final theorem returns only `[propext, Classical.choice, Quot.sound]`. We stress what this does and does not mean: it means the *stated* theorems are correct relative to Lean's kernel and Mathlib; it does *not* certify the recurrence for general n, nor any of the conjectural material. We give the human proofs in full here precisely so that they can be checked by a mathematician without trusting the machine. We would be grateful for scrutiny of both.

Calibrated significance, and a request for expert review. We want to state plainly how significant we believe this work to be, so as not to waste a reader's time. The sequence S_{20} is one member of a family that has been searched systematically (by Zagier, Almkvist–Zudilin, Cooper, and others); a previously-unrecorded member of such a family is, at best, a modest addition — potentially a new entry in the spirit of the Almkvist–van Straten–Zudilin tables of Calabi–Yau equations — and not a breakthrough. We are aware, too, that the cubic supercongruence of §3, while pleasant to formalize, is *elementary*: it holds for any sum $\sum_k \binom{n}{k}^a \binom{n+k}{k}$ with $a \geq 3$, for the same reason (Remark in §3). We therefore make no claim of novelty or importance beyond "we did not find this exact sequence previously recorded, and here is what we can and cannot prove about it." The genuinely interesting questions — a certified order-4 Picard–Fuchs recurrence, a correct identification of the underlying geometry, modularity, and a proof of the conjectural mod-p^3 refinement — are exactly those we leave open (§8). We would be especially grateful to specialists in Apéry-like sequences and in Calabi–Yau differential operators for a skeptical reading: in particular, for a prior appearance of S_{20} or its operator (which we would gladly acknowledge), for whether the order-4 operator is genuinely new, and for the correct statement of the associated geometry.

2 Preliminaries

Throughout, p denotes a prime and $\binom{n}{k}$ the usual binomial coefficient, with $\binom{n}{k} = 0$ for $k > n$. We write the summand of (1) as

$$T(n,k) \;=\; \binom{n}{k}^4 \binom{n+k}{k}, \qquad S_{20}(n) \;=\; \sum_{k=0}^{n} T(n,k).$$

We use two elementary facts repeatedly.

Lemma 2.1 (Prime divides interior binomials)**.** *If p is prime and $1 \leq k \leq p-1$, then $p \mid \binom{p}{k}$.*

Proof. From $k\binom{p}{k} = p\binom{p-1}{k-1}$ we get $p \mid k\binom{p}{k}$. Since $1 \leq k \leq p-1$, the prime p does not divide k, hence $p \mid \binom{p}{k}$. (This is `Nat.Prime.dvd_choose_self` in Mathlib.) □

Lemma 2.2 (Factorial p-divisibility)**.** *For a prime p and $m \in \mathbb{N}$, one has $p \mid m!$ if and only if $p \leq m$.*

Proof. If $p \leq m$ then p is one of the factors of $m!$. Conversely, if $p \mid m!$ then, as p is prime, p divides one of $1, 2, \ldots, m$, forcing $p \leq m$. (This is `Nat.Prime.dvd_factorial`.) □

3 The cubic supercongruence $S_{20}(p) \equiv 3 \pmod{p^3}$

The main arithmetic result of this note is the following.

Theorem 3.1 (Cubic supercongruence)**.** *For every prime $p \geq 5$,*

$$S_{20}(p) \;\equiv\; 3 \pmod{p^3}.$$

The proof has two ingredients: an elementary *collapse* of the sum to its two boundary terms modulo p^3 (Proposition 3.2), and the classical Wolstenholme congruence for $\binom{2p}{p}$ (Theorem 4.5, proved independently in §4). The collapse is the part specific to S_{20}, and it is genuinely simple; the arithmetic depth is entirely in Wolstenholme.

3.1 The collapse to boundary terms

Proposition 3.2 (Collapse)**.** *For every prime p,*

$$S_{20}(p) \;\equiv\; 1 + \binom{2p}{p} \pmod{p^3}.$$

Proof. Split the sum (1) at its endpoints:

$$S_{20}(p) \;=\; T(p,0) \;+\; \sum_{k=1}^{p-1} T(p,k) \;+\; T(p,p).$$

The boundary terms are explicit:

$$T(p,0) = \binom{p}{0}^4 \binom{p}{0} = 1, \qquad T(p,p) = \binom{p}{p}^4 \binom{2p}{p} = \binom{2p}{p}.$$

It remains to show that each interior term vanishes modulo p^3. Fix $1 \le k \le p-1$. By Lemma 2.1, $p \mid \binom{p}{k}$, hence $p^4 \mid \binom{p}{k}^4$. The companion factor $\binom{p+k}{k}$ is an integer, so

$$p^4 \;\Big|\; \binom{p}{k}^4 \binom{p+k}{k} = T(p,k),$$

and a fortiori $p^3 \mid T(p,k)$. Summing, $p^3 \mid \sum_{k=1}^{p-1} T(p,k)$, which gives the claim. □

Remark 3.3. The factor of $\binom{p}{k}^4$ is crucial: a single power would give only $p^1 \mid \binom{p}{k}$, enough for a mod-p statement but not mod-p^3. The fourth power provides p^4 "for free," so the companion binomial $\binom{p+k}{k}$ — whatever its p-adic valuation — cannot spoil divisibility by p^3. This is the only place the exponent 4 in (1) is used in this section.

3.2 Proof of Theorem 3.1

Proof. By Proposition 3.2, $S_{20}(p) \equiv 1 + \binom{2p}{p} \pmod{p^3}$. By Wolstenholme's theorem (Theorem 4.5), valid for $p \ge 5$, we have $\binom{2p}{p} \equiv 2 \pmod{p^3}$. Therefore

$$S_{20}(p) \;\equiv\; 1 + 2 \;=\; 3 \pmod{p^3}.$$ □

Corollary 3.4 (Unconditional mod p^2)**.** *For* every *prime p, $S_{20}(p) \equiv 3 \pmod{p^2}$.*

Proof. The collapse (Proposition 3.2) holds mod p^3 and hence mod p^2. Babbage's congruence $\binom{2p}{p} \equiv 2 \pmod{p^2}$ holds for all primes (Theorem 4.2), so $S_{20}(p) \equiv 1 + 2 = 3 \pmod{p^2}$. No lower bound on p is needed here. □

4 Wolstenholme's theorem from first principles

For completeness and because our Lean development does not assume it, we prove the two congruences for the central binomial coefficient used above. Nothing here is new — these are classical results of Babbage (1819) and Wolstenholme (1862) — but we give self-contained arguments matching the formalization.

The starting point is a Vandermonde identity.

Lemma 4.1 (Central binomial as a sum of squares)**.** *For all $p \ge 0$,* $\displaystyle\binom{2p}{p} = \sum_{k=0}^{p} \binom{p}{k}^2.$

Proof. Chu–Vandermonde gives $\binom{2p}{p} = \sum_{k=0}^{p} \binom{p}{k}\binom{p}{p-k}$, and $\binom{p}{p-k} = \binom{p}{k}$. □

Theorem 4.2 (Babbage, mod p^2)**.** *For every prime p, $\binom{2p}{p} \equiv 2 \pmod{p^2}$.*

Proof. By Lemma 4.1, $\binom{2p}{p} = \binom{p}{0}^2 + \sum_{k=1}^{p-1}\binom{p}{k}^2 + \binom{p}{p}^2 = 2 + \sum_{k=1}^{p-1}\binom{p}{k}^2$. For $1 \le k \le p-1$, Lemma 2.1 gives $p \mid \binom{p}{k}$, hence $p^2 \mid \binom{p}{k}^2$. Thus the interior sum is divisible by p^2 and $\binom{2p}{p} \equiv 2 \pmod{p^2}$. □

To upgrade to mod p^3 we need the harmonic input. Work in the field $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$.

Lemma 4.3 (Pascal residues)**.** *For a prime p and $0 \le j \le p-1$, $\binom{p-1}{j} \equiv (-1)^j \pmod{p}$.*

Proof. Induction on j. For $j = 0$, $\binom{p-1}{0} = 1$. For the step, Pascal's rule gives $\binom{p}{j+1} = \binom{p-1}{j} + \binom{p-1}{j+1}$. By Lemma 2.1, $p \mid \binom{p}{j+1}$ when $1 \le j+1 \le p-1$, so in $\mathbb{F}_p$ we have $\binom{p-1}{j+1} \equiv -\binom{p-1}{j} \equiv -(-1)^j = (-1)^{j+1}$. □

Lemma 4.4 (Harmonic core)**.** *For a prime $p \ge 5$, $\displaystyle\sum_{k=1}^{p-1} \frac{1}{k^2} \equiv 0 \pmod{p}$, where $1/k$ denotes the inverse of k in $\mathbb{F}_p$.*

Proof. As k ranges over $1, \dots, p-1$, the inverses k^{-1} range over the same set $\mathbb{F}_p^\times$, so $\sum_{k=1}^{p-1} k^{-2} = \sum_{x \in \mathbb{F}_p^\times} x^2 = \sum_{x \in \mathbb{F}_p} x^2$ (the $x = 0$ term is 0). For any finite field $\mathbb{F}_q$ and exponent i with $0 < i < q-1$, one has $\sum_{x \in \mathbb{F}_q} x^i = 0$ (a standard fact: the sum of all i-th powers over a finite field vanishes unless $(q-1) \mid i$). Here $q = p$, $i = 2$, and $2 < p-1$ exactly when $p \ge 5$, so the sum is 0. □

Theorem 4.5 (Wolstenholme, mod p^3)**.** *For every prime $p \ge 5$, $\binom{2p}{p} \equiv 2 \pmod{p^3}$.*

Proof. By Lemma 4.1 and the boundary evaluation as in Theorem 4.2,

$$\binom{2p}{p} - 2 \;=\; \sum_{k=1}^{p-1} \binom{p}{k}^2.$$

For $1 \le k \le p-1$ write $\binom{p}{k} = \frac{p}{k}\binom{p-1}{k-1}$, which is the integer identity $k\binom{p}{k} = p\binom{p-1}{k-1}$ solved for $\binom{p}{k}$; set $d_k := \binom{p}{k}/p = \frac{1}{k}\binom{p-1}{k-1} \in \mathbb{Z}$ (an integer by Lemma 2.1). Then $\binom{p}{k}^2 = p^2 d_k^2$ and

$$\binom{2p}{p} - 2 \;=\; p^2 \sum_{k=1}^{p-1} d_k^2.$$

Thus $\binom{2p}{p} \equiv 2 \pmod{p^3}$ if and only if $p \mid \sum_{k=1}^{p-1} d_k^2$. Reduce d_k modulo p. From $k\,d_k = \binom{p-1}{k-1}$ and Lemma 4.3, $k\,d_k \equiv (-1)^{k-1} \pmod{p}$, so in $\mathbb{F}_p$

$$d_k \equiv (-1)^{k-1} k^{-1}, \qquad d_k^2 \equiv k^{-2}.$$

Hence $\sum_{k=1}^{p-1} d_k^2 \equiv \sum_{k=1}^{p-1} k^{-2} \equiv 0 \pmod{p}$ by Lemma 4.4 (using $p \ge 5$). Therefore $p \mid \sum_k d_k^2$ and $\binom{2p}{p} \equiv 2 \pmod{p^3}$. □

Remark 4.6. The hypothesis $p \ge 5$ enters in exactly one place: Lemma 4.4 needs $2 < p-1$. For $p = 3$ one checks directly $\binom{6}{3} = 20 \equiv 2 \pmod{9}$ but $20 \equiv 20 \not\equiv 2 \pmod{27}$, so the mod-$p^3$ statement genuinely fails at $p = 3$; the mod-p^2 statement (Theorem 4.2) survives.

5 The Apéry-style boundary congruence

The second sequence-specific result concerns the value at $p-1$.

Theorem 5.1 (Apéry-style congruence mod p)**.** *For every prime p, $S_{20}(p-1) \equiv 1 \pmod{p}$.*

The mechanism mirrors the collapse of §3, but now the divisibility comes from the *companion* binomial rather than from $\binom{p-1}{k}$.

Lemma 5.2 (Companion divisibility)**.** *For a prime p and $1 \le k \le p-1$, $p \mid \binom{p-1+k}{k}$.*

Proof. The factorial identity for binomial coefficients gives

$$\binom{p-1+k}{k} \cdot k! \cdot (p-1)! \;=\; (p-1+k)!,$$

since $(p-1+k)-k = p-1$. As $1 \le k \le p-1$ we have $p \le p-1+k$, so by Lemma 2.2 the prime p divides the right-hand side $(p-1+k)!$. On the other hand $k < p$ and $p-1 < p$, so by Lemma 2.2 again $p \nmid k!$ and $p \nmid (p-1)!$. Since p is prime and divides the product $\binom{p-1+k}{k}\, k!\,(p-1)!$ but neither $k!$ nor $(p-1)!$, it must divide the remaining factor $\binom{p-1+k}{k}$. □

Proof of Theorem 5.1. Split the sum at $k=0$:

$$S_{20}(p-1) \;=\; T(p-1,0) \;+\; \sum_{k=1}^{p-1} T(p-1,k), \qquad T(p-1,0) = \binom{p-1}{0}^4 \binom{p-1}{0} = 1.$$

For each interior index $1 \le k \le p-1$, $T(p-1,k) = \binom{p-1}{k}^4\binom{p-1+k}{k}$, and Lemma 5.2 gives $p \mid \binom{p-1+k}{k}$, hence $p \mid T(p-1,k)$. Therefore $\sum_{k=1}^{p-1} T(p-1,k) \equiv 0 \pmod{p}$ and $S_{20}(p-1) \equiv 1 \pmod{p}$. □

Conjecture 5.3 (Strong Apéry-style congruence)**.** For every prime $p \ge 5$, $S_{20}(p-1) \equiv 1 \pmod{p^3}$.

We have verified Conjecture 5.3 for all primes $5 \le p \le 200$ (see the reproducibility report). We do *not* have a proof; the mod-p argument above does not extend naively, because $\binom{p-1}{k}^4 \equiv 1 \pmod{p}$ controls only the leading p-adic order. A proof would presumably require a harmonic-sum analysis in the spirit of §4 (likely involving $\sum 1/k$ and $\sum 1/k^2$ to higher order), and we would welcome one.

6 Numerical and formal verification

All numerical claims are reproduced by a single self-contained script, `python/verify_all.py` (standard library only), whose checks include: the first 18 terms against an independent transcription; the cubic supercongruence and Wolstenholme/Babbage congruences for all primes up to 200; the Apéry-style congruence mod p (and the conjectural mod p^3) up to 200; the collapse identity of Proposition 3.2; and the diagonal evaluation discussed in §8. The script terminates with exit code 0 and the message *"ALL NUMERICAL CHECKS PASSED."*

Theorems 3.1, 4.5 (with Theorem 4.2), and 5.1 have been formalized in Lean 4 (Mathlib, toolchain `v4.31.0`) as, respectively, `Supercongruence.supercongruence_unconditional`, `Wolstenholme.wolstenholme` / `Wolstenholme.babbage`, and `AperyCongruence.apery_congruence`. Each compiles under `lake build` and, under `#print axioms`, depends only on `propext`, `Classical.choice`, and `Quot.sound`. The full audit is recorded in `VERIFICATION_REPORT.md`.

7 The Picard–Fuchs operator: order and Calabi–Yau structure

This section reports the computational study of the operator governing S_{20}. We are careful to separate what is *proved* (the recurrence, via a certificate), what is *exactly computed* (the differential order and the local exponents), and what remains *conjectural* (the Calabi–Yau identification). The computations are reproducible from the scripts in `src/picard_fuchs/` of the repository.

7.1 The minimal recurrence has order 4, and is proved for all n

Proposition 7.1 (Minimal P-recurrence)**.** *The sequence $S_{20}(n)$ satisfies a linear recurrence*

$$\sum_{j=0}^{4} P_j(n)\, S_{20}(n+j) = 0,$$

with explicit integer polynomials $P_0, \ldots, P_4$ of degree 13, and no such recurrence of order 2 or 3 exists.

Evidence and proof. The order and the exact coefficients were obtained *four independent ways*, which agree coefficient-for-coefficient (up to a global sign): (i) an exact nullspace over $\mathrm{GF}(p)$ for two primes, heavily over-determined; (ii) an exact nullspace over $\mathbb{Q}$, verified on 101 consecutive terms; (iii) `ore_algebra`'s `guess` in SageMath; and (iv) Maxima's `Zeilberger`. The non-existence of an order-2 or order-3 relation was checked over the same range. Crucially, Maxima's `Zeilberger` returns the order-4 telescoper *together with a rational certificate* $R(n,k)$ satisfying

$$\sum_{j=0}^{4} P_j(n)\, T(n+j,k) \;=\; \Delta_k\big[R(n,k)\, T(n,k)\big], \qquad \Delta_k g(k) := g(k+1) - g(k),$$

for the summand $T(n,k) = \binom{n}{k}^4 \binom{n+k}{k}$. Summing over k telescopes the right-hand side to boundary terms, which vanish; hence the recurrence holds *for all n*, not merely on the tested range. The leading coefficient returned by Maxima equals our $P_4(n) = (n+3)^2(n+4)^4\, Q(n)$ with Q an explicit degree-7 integer polynomial. □

Remark 7.2. The certificate constitutes a proof in the standard sense of creative telescoping. A fully kernel-checked version — re-verifying the (finite, denominator-cleared) rational identity inside Lean 4 — is not yet done; we flag it as the natural next step and an inviting, self-contained formalization target.

7.2 The differential operator has order 6, with an order-4 MUM block

It is tempting to read the recurrence order (4) as the order of the Picard–Fuchs operator, but these are different invariants. Let $f(z) = \sum_{n\geq 0} S_{20}(n)\, z^n$.

Proposition 7.3 (Holonomic rank)**.** *The minimal linear ODE with polynomial coefficients annihilating $f(z)$ has order 6 (and coefficient degree 15); no annihilating operator of order ≤ 5 exists.*

This was established by an exact nullspace computation on the Taylor coefficients (two primes, stable under heavy over-determination), and agrees with `ore_algebra`'s series `guess`. The apparent tension with Proposition 7.1 is resolved by the local analysis at the origin.

Proposition 7.4 (Indicial equation at $z = 0$)**.** *Writing the order-6 operator in the form $\sum_{i=0}^{6} R_i(z)\, \theta^i$ with $\theta = z\,\mathrm{d}/\mathrm{d}z$, its indicial polynomial at $z = 0$ is*

$$-715\, s^4 (s-1)^2,$$

so the local exponents at $z = 0$ are $\{0, 0, 0, 0,\ 1, 1\}$.

The exponent 0 occurs with multiplicity 4: this is a *maximal-unipotent-monodromy* (MUM) block of order 4, the structural hallmark of a Calabi–Yau 3-fold Picard–Fuchs operator (the situation of the Apéry numbers for $\zeta(3)$). The remaining exponents $\{1, 1\}$ form an integer-shifted order-2 block, the signature of an *apparent singularity*: it raises the holonomic rank of f from 4 to 6 but carries no new transcendental content. Thus the geometrically essential operator is the order-4 MUM block, consistent with Proposition 7.1.

Remark 7.5 (Honest status of the factorization). We have not yet *exhibited* the factorization $L_6 = L_4 \cdot L_2$ with L_4 irreducible; the local-exponent data strongly indicates it, but a rigorous factorization (and the irreducibility of L_4) requires a version-matched computer-algebra setup and would benefit from expert verification.

7.3 Mirror map and instanton numbers

Computing the holomorphic period f and the logarithmic period $g(z) = \sum_n B(n)z^n$ with $B(n) = \sum_k \binom{n}{k}^4 \binom{n+k}{k} \left[4(H_n - H_{n-k}) + (H_{n+k} - H_n)\right]$, the mirror map $q(z) = z \exp(g/f)$ has *integer* coefficients $q_1 = 1, q_2 = 9, q_3 = 165, q_4 = 4110, \ldots$ for $1 \leq d \leq 16$ (exact rational arithmetic). This is consistent with Lian–Yau integrality and independently corroborates the MUM structure.

We must, however, report a genuine *negative*. The more stringent test is integrality of the genus-0 instanton numbers n_d extracted from the normalized Yukawa coupling $K(q) = n_0 + \sum_d n_d \, d^3 q^d/(1 - q^d)$. With a *placeholder* normalization (a second θ_q-derivative of the mirror map) the resulting n_d are *not* integers, and their denominators are exactly of the form d^3 (e.g. $165/2^3$, $3463611/5^3$, $2144305763/7^3$). This denominator pattern is the signature of an incorrect Yukawa normalization rather than a failure of Calabi–Yau structure: the geometrically correct coupling must be read off the (desingularized) order-4 operator L_4, which we have not yet isolated. We therefore regard instanton integrality as *unresolved*, and the mirror-map integrality above as the robust positive evidence.

7.4 Summary of the geometric picture

The data support, but do not prove, the following picture: S_{20} is the holomorphic period of a one-parameter family of Calabi–Yau 3-*folds*; the associated operator is the order-4 MUM block sitting inside the order-6 holonomic operator of f. In particular the earlier "weight-5 / Calabi–Yau 4-fold" description (where "weight 5" was merely the count of binomial factors) is *incorrect*: the operator-theoretic invariant points to a 3-fold. We state this as a corrected *conjecture*, contingent on the factorization and the correct instanton normalization.

8 What remains open

We close by stating clearly the parts of the broader picture that are *not* theorems in this paper, in the hope that experts will engage with them.

Operator factorization and a Lean-checked certificate. Two concrete items from §7 remain open: a rigorous factorization $L_6 = L_4 \cdot L_2$ exhibiting the irreducible order-4 Calabi–Yau operator L_4, and a kernel-checked (Lean 4) re-verification of the Zeilberger certificate's finite rational identity. Both are self-contained and would materially strengthen the formal status of the results.

Diagonal representation. We observe the identity, verified for $0 \le n \le 6$ and provable by elementary power-series extraction,

$$S_{20}(n) \;=\; \mathrm{Diag}\left[\frac{1}{(1-x_1)(1-x_2)(1-x_3)(1-x_4)(1-x_5) - x_1x_2x_3x_4}\right],$$

i.e. $S_{20}(n)$ is the coefficient of $(x_1 \cdots x_5)^n$. Indeed expanding $1/(D-M) = \sum_{r\ge0} M^r/D^{r+1}$ with $D = \prod_i (1-x_i)$, $M = x_1x_2x_3x_4$ and extracting the diagonal yields exactly $\sum_r \binom{n}{r}^4 \binom{n+r}{r}$. The *symmetric* five-variable candidate does not represent S_{20}; its diagonal is 2^n.

Calabi–Yau period and instanton normalization. As detailed in §7, the order-4 MUM block and the integral mirror map are evidence — not proof — that S_{20} is a Calabi–Yau 3-fold period. The correct Yukawa normalization (and hence a genuine instanton-number integrality test) requires the isolated operator L_4 and deserves expert scrutiny; the precise geometry remains conjectural.

Invitation. This is a preprint released specifically to invite review. Corrections, counterexamples, a proof of Conjecture 5.3, a verified recurrence certificate, an identification (or refutation) of the Calabi–Yau interpretation, or a prior appearance of S_{20} in the literature would all be very welcome. Please open an issue or pull request at the repository.

Acknowledgements

We thank the OEIS editors and the Lean/Mathlib community. Any errors are the author's own, and the work is offered in a spirit of provisional, correctable inquiry rather than finality.

A An applied side-quest, reported in full: a learnable positional bias inspired by—but not explained by—the Calabi–Yau structure

We record here an engineering experiment that the holonomic structure of S_{20} *inspired*, together with its honest—and partly cautionary—outcome. It is not part of the mathematical contribution; we include it because the negative and null results are as informative as the positive one, and because reporting them is the discipline this preprint tries to model.

Hypothesis. Accelerators are memory-bandwidth bound. A positional attention bias generated *on the fly* from the proved order-4 recurrence of S_{20} would cost $\mathcal{O}(L)$ memory rather than an $\mathcal{O}(L^2)$ table—*if* it preserves model quality. We fixed, in advance, a quality gate: PASS within +1% of the best baseline perplexity, KILL at $> 5\%$ regression (all on WikiText-2, models trained from scratch, one per positional scheme, at matched compute—the ALiBi-paper methodology).

Outcome, in order.

1. A Triton kernel reproduces the reference attention to FP16 tolerance (correctness gate PASS).
2. The *fixed* bias $-\log S_{20}(d)$ **fails** the quality gate (+10.2% perplexity; a plain sliding window wins). Its slope $\log\lambda \approx 3.76$ is too steep—a brick wall.

3. Making the per-head slope *learnable*, $\text{bias}_h(d) = -\gamma_h \log S_{20}(d)$ with γ_h trained, **passes at scale**: +8.1% better perplexity than the best baseline (3.32 vs 3.62), with stable length-extrapolation where learned absolute positions collapse. (An intermediate run taught a methodological lesson: a short training budget crowned the high-capacity bias, which then overfit at full budget; γ-regularization and validation early-stopping resolve it.)

4. **Attribution.** A nested bias $\text{bias}_h(d) = -a_h\, d + b_h \log d$ (slope a and log-curvature b both learnable per head) shows the gain comes from *learnability*, not the Calabi–Yau shape: a learnable ALiBi slope with $b \equiv 0$ already beats the fixed holonomic bias, and when b is free the optimizer drives it *away* from the geometric value $\beta = 2$. We therefore drop the S_{20}/recurrence machinery and report the result as what it is—a learnable linear-plus-log positional bias—rather than as a number-theoretic method.

5. **Final form selected by elimination.** Three independent GPU bake-offs converge on the simplest scheme. A six-family comparison (linear, log-curvature, Fourier, fixed Calabi–Yau, local+tail hybrid, exponential decay) leaves *learnable per-head ALiBi* the winner; everything more expressive loses at scale. A final orthogonal probe of per-head content/position balance—a learnable scale c_h on the content score, optionally with a per-head softmax temperature—only *ties* learnable-ALiBi (+0.35% at 4× the training length), while the temperature knob *regresses*. The shipped hypothesis is thus one learnable linear slope per head and nothing else: a single per-head scalar, $\mathcal{O}(1)$ state, no curvature, no sequence, no content scale. *Adaptability beats added structure—up to that one slope.*

What we claim, and what we do not. We claim a small, reproducible quality result (a learnable per-head positional bias competitive with, and on this benchmark a few percent better than, ALiBi, with good extrapolation). We do *not* claim that the Calabi–Yau geometry improves attention (the attribution refutes it), nor a memory advantage over FlashAttention+ALiBi (which already generates its bias on the fly; a learnable per-head scalar is $\mathcal{O}(1)$ state, identical to ALiBi). The $\mathcal{O}(L)$-vs-$\mathcal{O}(L^2)$ saving holds only against a *materialized* bias table, a strawman in modern kernels. Full data and code are in the repository (`docs/CYSIEVE_JOURNEY.md`) and an open benchmark dataset; we would value a fair comparison against learnable-ALiBi, NoPE, and RoPE-scaling on a genuine long-range task.

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